

HTML: HyperText Markup Language – a hypertext markup language invented by Tim Berners-Lee and becoming the standard language for creating web pages.

HTML uses tags to format data and create frames/tables for web pages.

Basic components (tag, element, property)

Basic HTML tags.

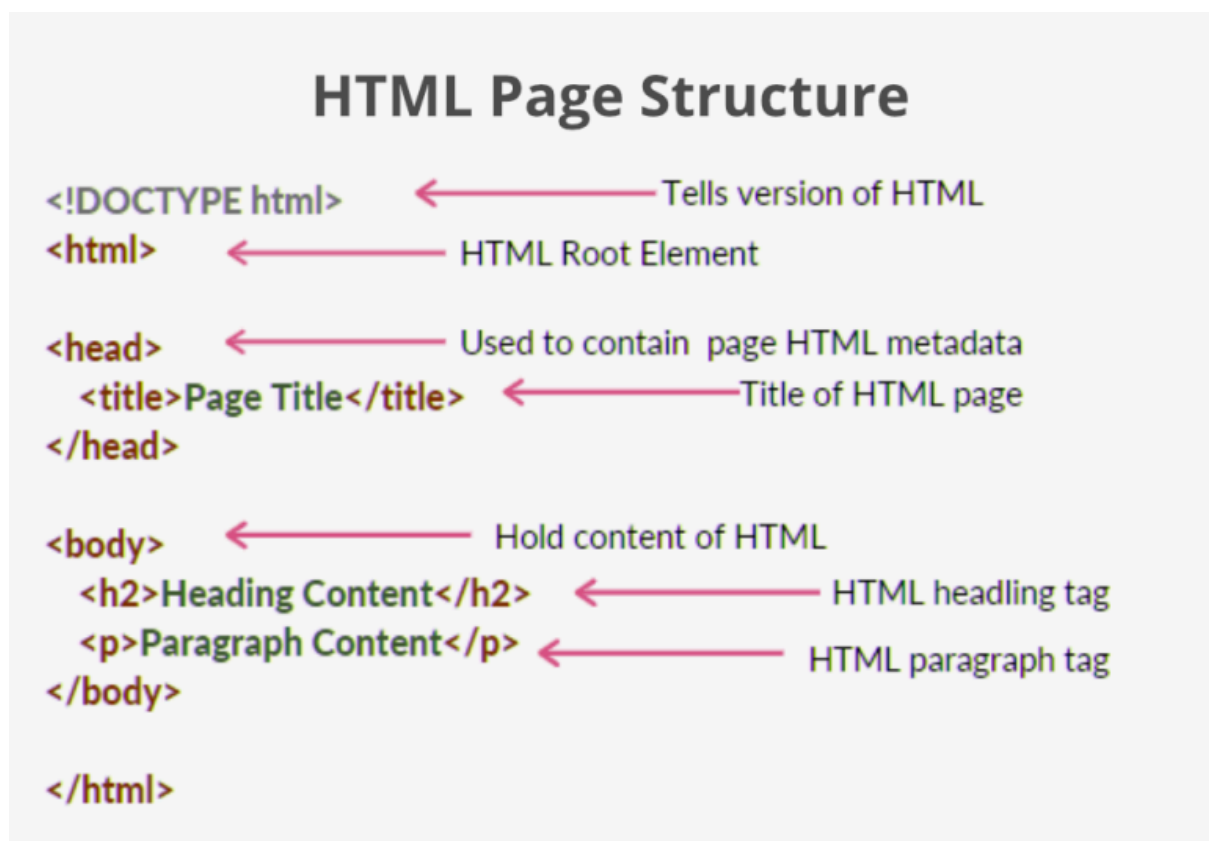
Page formatting tags: `<body>`

Text formatting tags: `<font>`, `<p>`, `<b>`, `<i>`, `<in>`...

Hyperlink tag: `<a>`

List formatting tags: `<ul>`, `<ol>`, `<li>`

Image insertion card: `<img>`



Tags

Tags are a set of symbols defined in HTML that have a special meaning.

- **Structure:** Tags starting with a `<` and ends with a `>`.
- **Opening card:** `<tag name>`
- **Closed card:** `</tag name>`

2. Element

An element can include an opening tag, a closing tag, and content inside that pair of tags. There are two main types of elements in HTML:

- **The element containing the content:** This includes both open and closed cards.
 - For example: `<p>Content</p>`
- **Empty element:** Includes only a single card (no content and no separate closing tag).
 - For example: `<br>` It can be written as `</br>` or `<br />`.

Concepts and Usage of Attributes

- **Define** Each HTML tag can have one or more attributes to provide additional information.
- **Location:** The attribute is entered immediately before the closing parenthesis `>` of the open card.
- **Quantity** A tag can contain multiple attributes, which are separated by spaces.
- **Rules of Value:**
 - The value of an attribute can be enclosed in double quotes `" "`, parentheses `' '` or without parentheses.
 - **Mandatory cases** If the value is a string containing spaces, a comma must be used. `" "` or `' '` to wrap the string.
- For example: `<table border="1" cellpadding="5">`.

Common attributes of cards `<body>`

Attributes	Meaning	For example
<code>bbgcolor="color"</code>	Set the background color for the website.	<code><body bgcolor="#00FF00"></code>
<code>background="url"</code>	Set the background image for the website.	<code><body background="image.jpg"></code>
<code>leftmargin="num"</code> <code>topmargin="num"</code>	Set the left and top margins for the webpage.	<code><body leftmargin="0" topmargin="0"></code>
<code>text="color"</code>	Set the text color for all text, including headings.	<code><body text="#FF0000"></code>

<pre> alink="color" vlink="color" link="color" </pre>	Define the colors for hyperlinks in the text, including: - alink: active link (currently being clicked) - vlink: visited link (already accessed) - link: link (not yet accessed)	<pre> <body alink="#FF0000" vlink="#00FF00" link="#0000FF"> </pre>
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Style formatting, font size, tag

– paragraph tag

This means that paragraphs are used to divide text into sections.

Welcome to HTML

HTML is a hypertext markup language.

Card

- New line Tag

Used to create a new line. For example:

Line 1

Line 2

Card - Bold text tag: This tag is used to bold a section of text. For example:

Welcome to HTML

The emphasis tag has a similar effect to the 'b' tag, meaning it will highlight and emphasize the text.

Welcome to HTML

Card `<p>`– paragraph card

Card `<p>`(Paragraph) is used to segment text.

For example:

<p>
Welcome to HTML
</p>
<p>
HTML is a hypertext markup language.
</p>

2. Card
– new line

Card
Used to create a new line.

For example:

<p>
Line 1
 Line 2
</p>

3. Card– in bold

Card Used to bold text.

For example:

<p>
Welcome to HTML
</p>

4. Card - emphasize

Card have a similar effectbut it carries a sense of emphasis.

For example:

<p>
Welcome to HTML
</p>

5. Card<i>– in italics

Card <i>(Italic) is used to italicize text.

For example:

```
<p>  
Welcome to <i>HTML</i>  
</p>
```

6. Card<mark> - tick

Card <mark>Used to highlight text.

For example:

```
<p>  
Welcome to <mark>HTML</mark>  
</p>
```

7. Card- middle line

Card Used to strike through text.

For example:

```
<p>  
Welcome to <del>HTML</del>  
</p>
```

8. Card<sub>- lower index

Card <sub>Used to display the subscript.

For example:

```
<p>  
H<sub>2</sub>SO<sub>4</sub>  
</p>
```

9. Card<sup>- upper index

Card <sup>Used to display the index above.

For example:

<p>
ax² + bx + c = 0
</p>

Comprehensive example

<body>
<h2>TEXT FORMATTING TAGS</h2>

My name is Nguyen Thi Huong.

This year I am 40 years old.

<p>PTB2 has the form: ax² + bx + c = 0</p>

<p>The chemical formula of water: H₂O</p>

<p>Unexpected <mark>discount</mark> campaign</p>

<p>HCEM: <u>Learning by Doing - Practical Training</u></p>

<p>Reduced from \$220 to \$199</p>

<p><center>HCEM</center></p>

</body>



HTML LAYOUT FORMATTING TAGS

1. Header

The header is the top part of a website.

It usually contains a logo, banner, and submenus such as information, a search box, dates, weather, etc.

2. Navigation

Navigation is the main menu of a website.

Typically displayed as a horizontal bar, containing the main items (main categories).

This section is very important because it helps to:

- Users can easily navigate between pages.
- Search engines understand website structure.

3. Content

Content refers to the main body of information on a website.

This area displays important information such as articles, products, news, etc.

4. Sidebar

A sidebar is the side panel of a website.

Typically contains:

- Side menu
- Related articles
- Advertisement

A website might have:

- 1 sidebar
- 2 sidebar
- Or more, depending on the design.

5. Footer

The footer is the bottom section (footer) of a website.

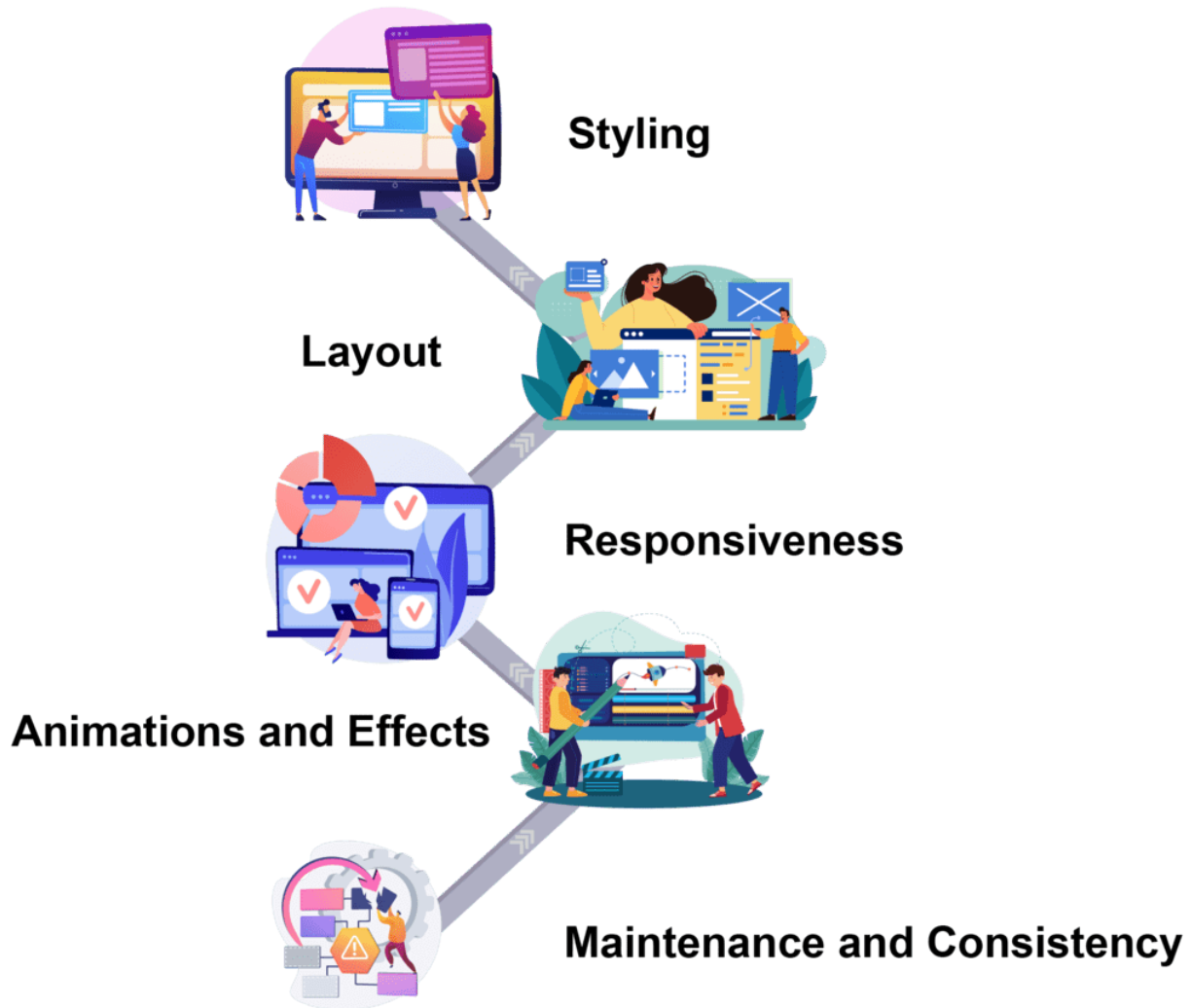
Typically contains:

- Copyright information
- Contact
- Terms of Use
- Secondary link

INTRODUCTION TO CSS



CSS used for?



CSS (Cascading Style Sheets) is a language used to define the presentation of documents such as HTML, XHTML, XML, SVG, etc.

HTML supports some basic formatting for text, images, tables, etc., but its formatting capabilities are limited and inconsistent across systems. CSS provides more presentation attributes, allowing for flexible and professional interface design.

Using direct formatting in HTML is time-consuming and increases file size. CSS, on the other hand, allows you to separate the interface elements into an external file, which helps:

- Easy to manage and edit
- Reuse for multiple pages
- Ensure consistency across the entire website.

For example, if you want to change the font or color for the entire website, you only need to edit the CSS file instead of modifying each HTML page individually.

2. CSS Syntax

Basic CSS syntax:

```
selector {  
    property: value;  
}
```

Components of syntax

Selector

This is the HTML object to which we want to apply formatting.
It could be:

- The HTML: body, h2, p, img
- id: #title, #content
- class: .username

Property

It is an attribute that specifies how it is presented, for example:

- color (text color)
- background-color
- font-family (font type)
- margin, padding (space)

Value

It is the value of an attribute, for example:

- color: red
- font-size: 16px

Notes when writing CSS

- Each attribute must have a value.
- If there are multiple attributes, separate them with a comma. ;

- The attributes are enclosed in brackets. `{ }`
- You can write multiple attributes to a selector.

For example:

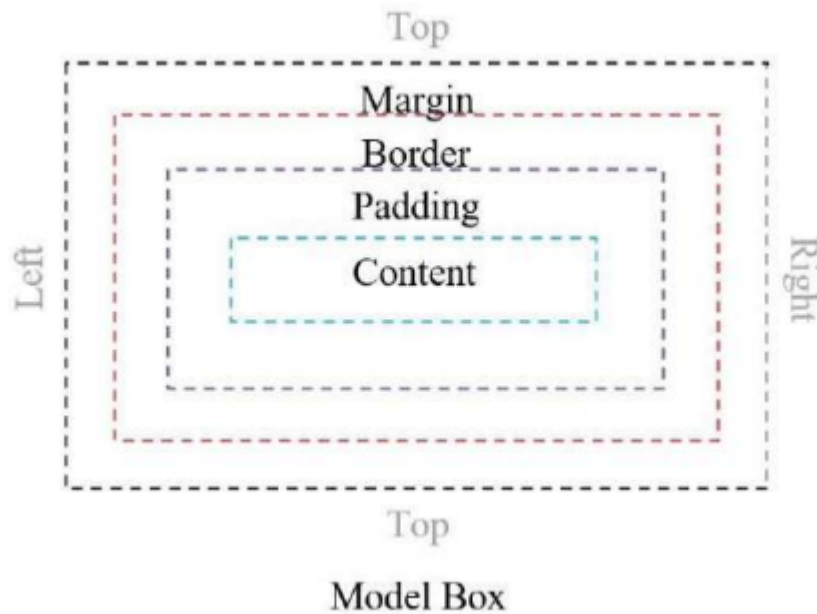
```
p {
  color: blue;
  font-size: 16px;
  margin: 10px;
}
```

+ Value: The value of the attribute. For example, as in the example above, the value is #FFF, used to set the background color of the page to white.

Color units in CSS

Unit	Describe
Color-name	Use color names in English. Some common examples include: black, white, red, green, blue, cyan, magenta,...
RGB (r, g, b)	The RGB color system uses three integer values for Red, Green, and Blue. Each value is in the range of 0 to 255. Their combination can create countless different colors.
RGB (%r, %g, %b)	Similar to the color system above, but instead of using integers, the R, G, and B values are determined as percentages. 0% to 100%.
Hexadecimal RGB	RGB color codes are written as hexadecimal system (starts with a mark #). For example: - #FFFFFF: white - #000000: black - #FF00FF: bright red (magenta)

CSS FORMATTING



CSS Text

color

Used to set the text color for an element.

For example: `color: red;`

text-align

Text alignment: left, right, center, justify.

letter-spacing

Adjust the spacing between characters.

text-decoration

Add text effects: underline, strikethrough, overline.

text-transform

Convert text: uppercase, lowercase, capitalize the first letter.

CSS Background

background-color

Set the background color for the element.

background-image

Insert an image as a background.

background-repeat

Specify background image repetition (horizontal, vertical, or no repetition).

background-attachment

Determine whether the background image scrolls with the page or is fixed.

background-position

Determine the display position of the background image.

background (abbreviated)

Combine multiple background attributes into a single line.

CSS Margin**margin**

Create outer spacing between the element and other elements.

Each direction can be written separately or combined.

CSS Padding**padding**

Create internal spacing between the content and the element's border.

CSS Border**border-width**

The thickness of the border.

border-style

Border styles: solid line, dashed line, double line, etc.

border-color

The color of the border.

border (shortened)

Combine border properties into a single line.

CSS Font**font-family**

Choose the font style to display.

font-style

Font style: normal, italic.

font-variant

Display text in small-caps format.

font-weight

The boldness of the text.

font-size

Font size.

font (abbreviated)

Combine multiple font attributes into a single line.